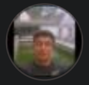
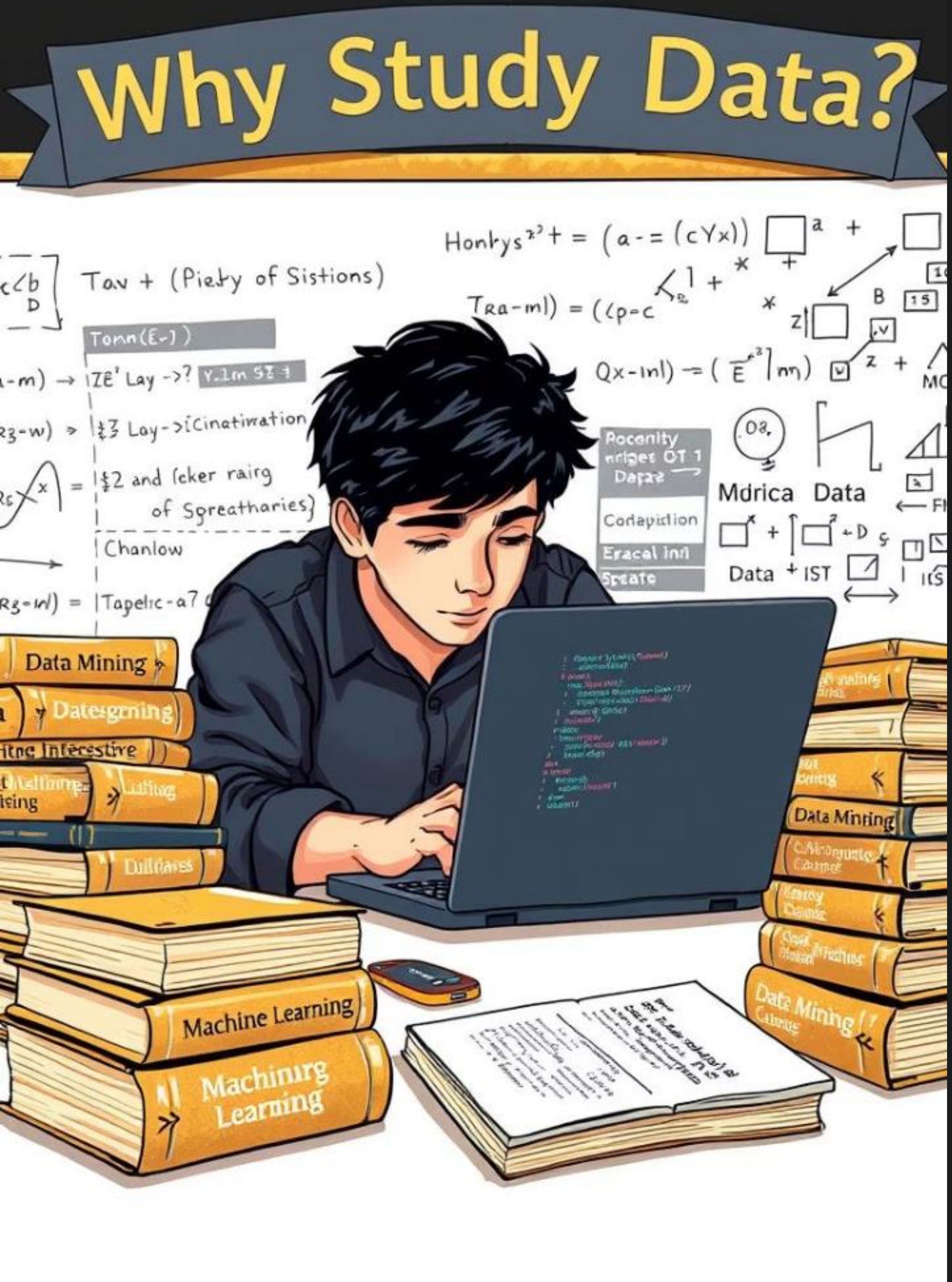
Interpretable Machine Learning for Youth Substance Use Analysis

Tree-Based Approaches on NSDUH 2023 Youth Data.

 **by Nikhil Ghugare**



Why Study Youth Substance Use?

1 Lifelong Impact

Influences lifelong behavioral and health outcomes.

2 Key Predictors

Peer behavior, parental support, school environment.

3 Model Aims

Identify risk, understand frequency, estimate days of use.

About the Dataset (NSDUH 2023 – Youth)

Sample Size

~10,500 youth responses under age 18.

Feature Groups

Substance Use, Demographics, Social Environment.

Data Addressed

Invalid/missing values, filtered or imputed codes.

From Data to Prediction: Our Our Process

1

Feature Engineering

Create new features from existing data.

2

Preprocessing

Handle missing values and balance classes.

3

Modeling

Apply Decision Tree Classifier & Regressor.

Who Has Ever Used Tobacco? Tobacco? (Binary)

Accuracy

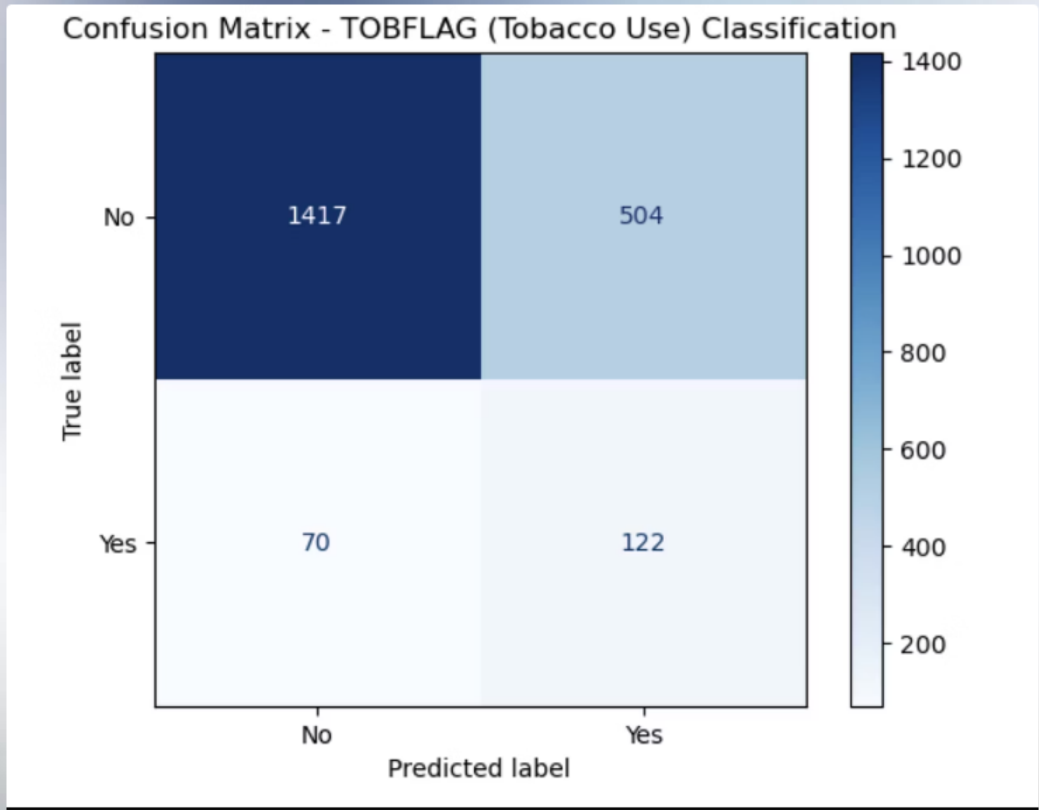
73% accuracy in predicting tobacco use.

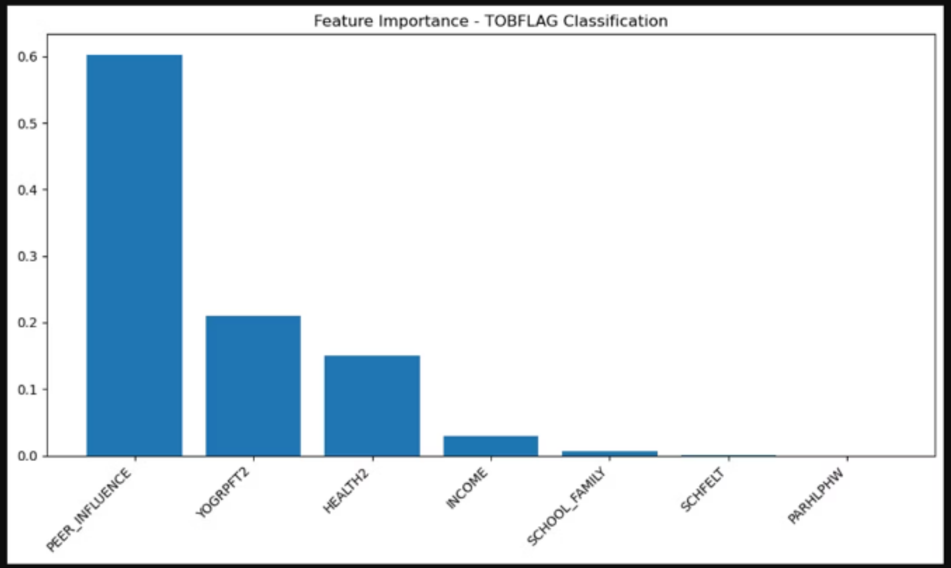
Top Features

Peer Influence, Income, Family Support.

Tree Info

55 nodes, 28 leaves, depth = 5.





How Frequently Is Smokeless Smokeless Tobacco Used? (Multi- Class)

5-Class

5-class classification of usage days.

Top Features

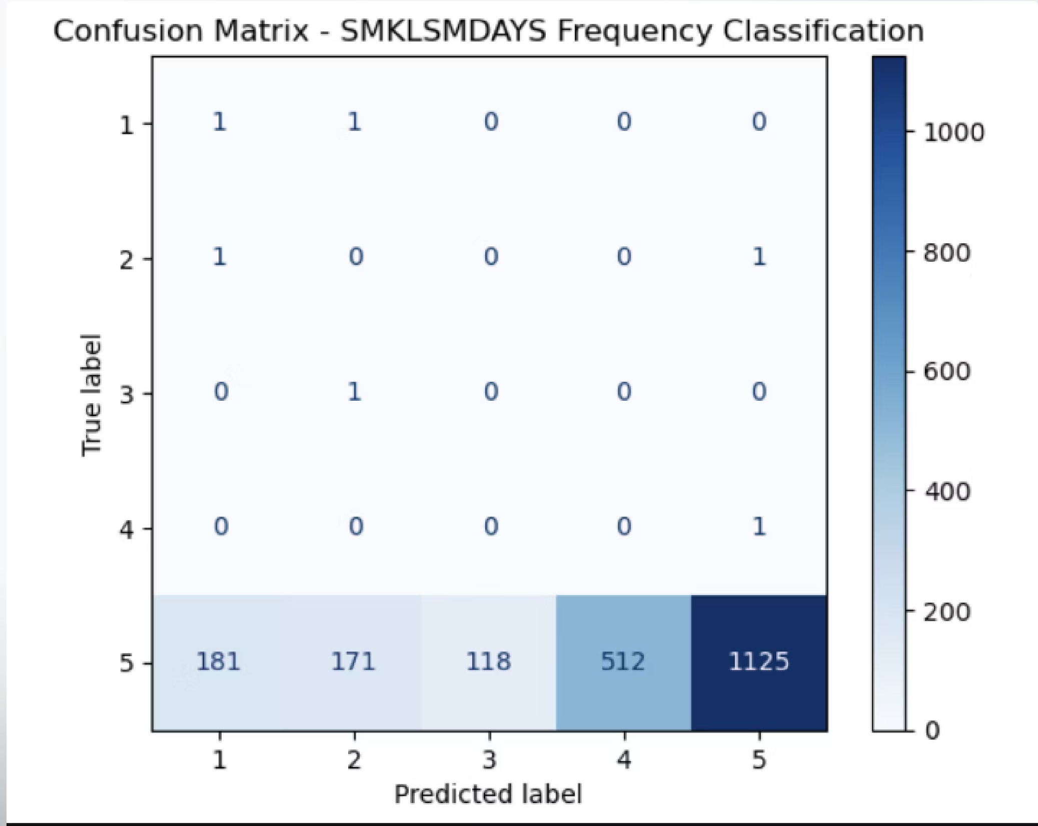
Health Status, Peer Influence, Income.

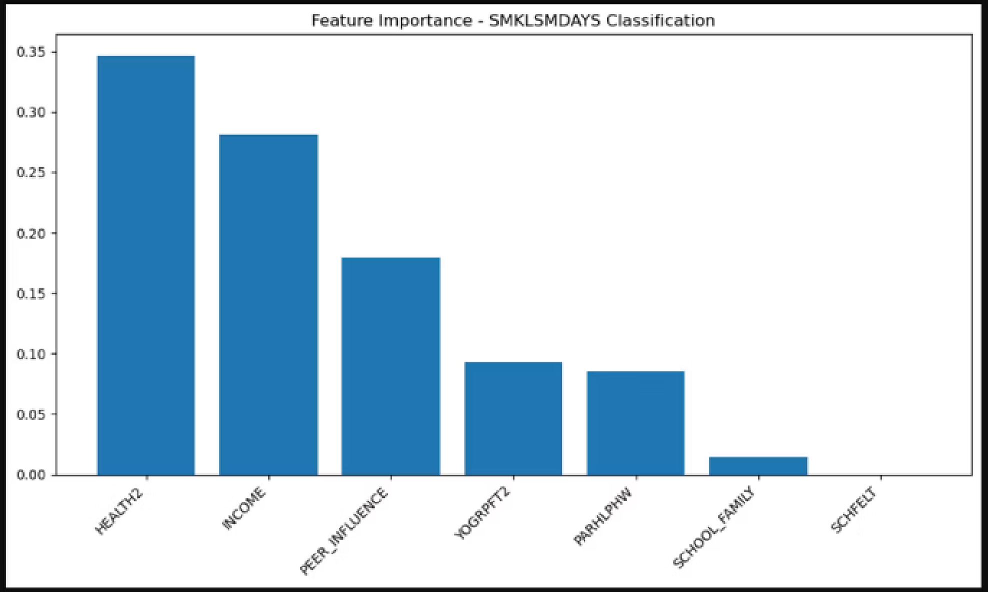
Accuracy

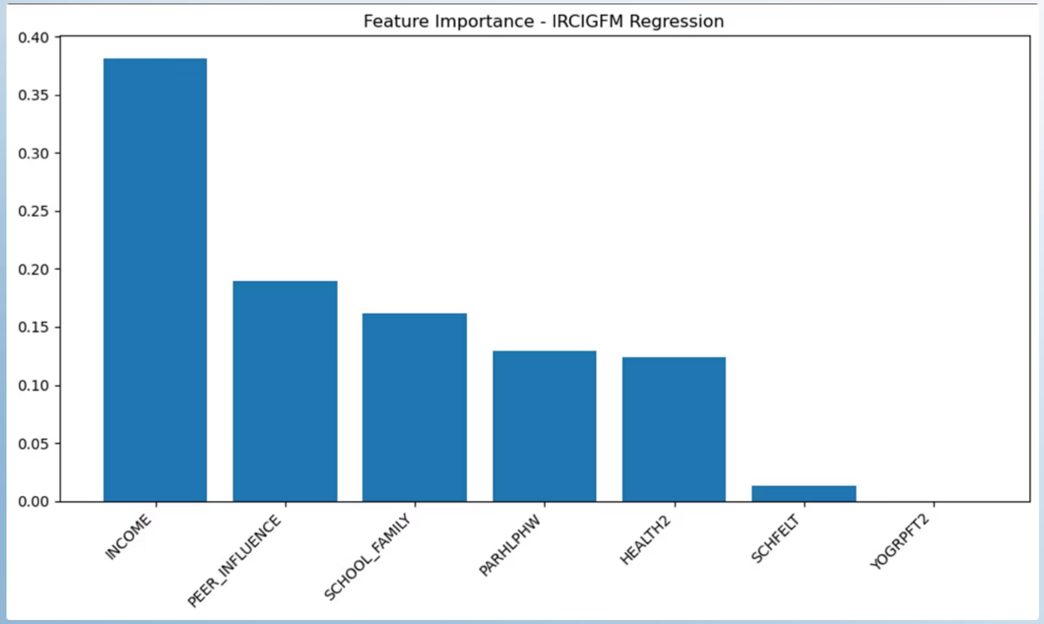
53% accuracy in predicting usage frequency.

Challenges

Class imbalance, sparse upper classes.







How Many Days Were Cigarettes Cigarettes Used? (Regression) (Regression)

-0.36

R^2 Score

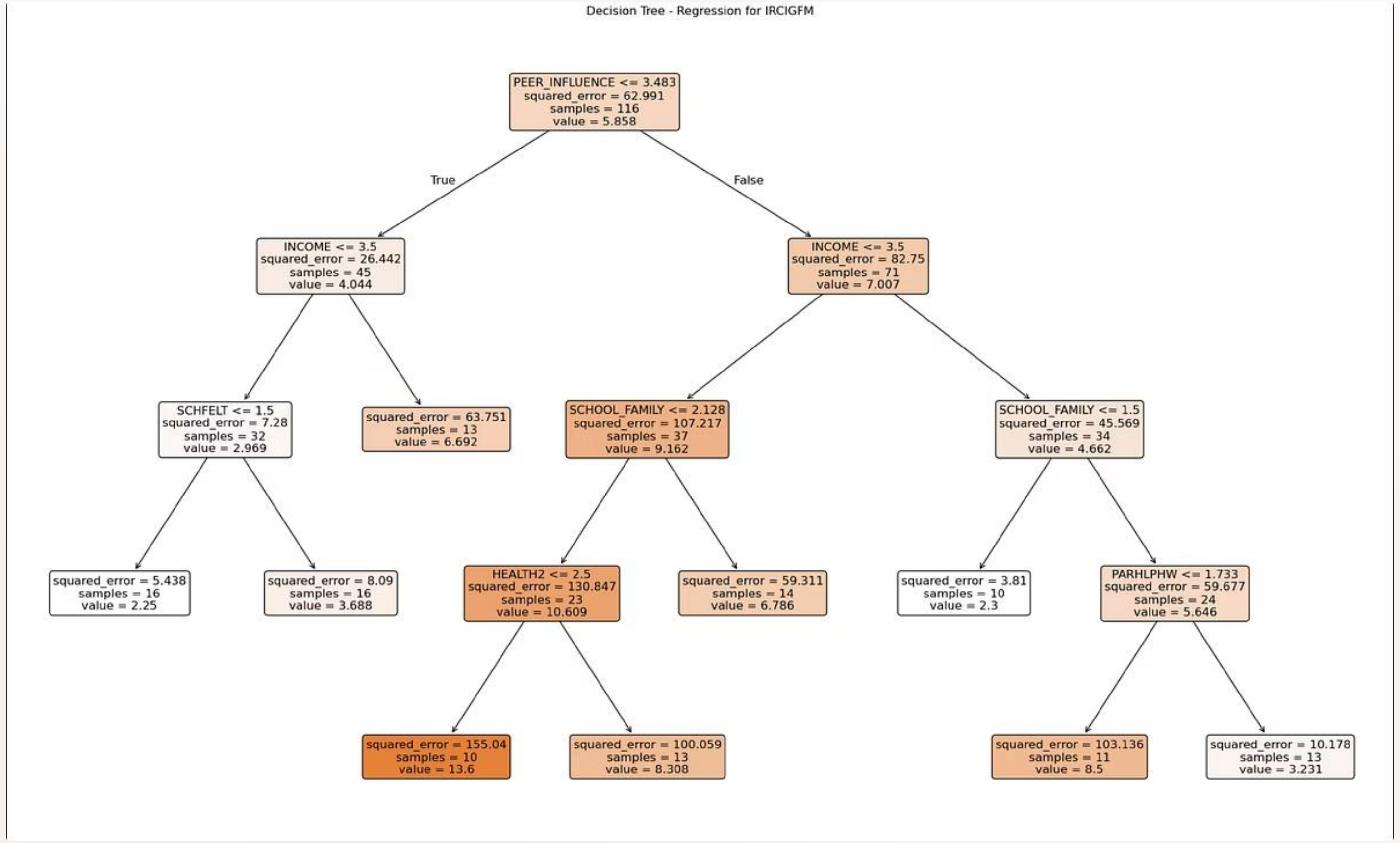
5.1

MAE (days)

7.0

RMSE (days)

Interpreting the Regression Tree – IRCIGFM



- **Goal**: Predict how many days a youth smoked in the past 30 days.
- **Root Split**: Lower **peer influence** leads to fewer smoking days.
- **Low Peer Influence** + **Low Income** + **Safe School** → ~2.2 days/month
- **High Peer Influence** + **Poor Health** + **Weak Family-School Support** → Up to ~13.6 days/month
- **Top Predictors**:
- PEER_INFLUENCE, INCOME, HEALTH2, SCHOOL_FAMILY

How Do Decision Trees Split?

1

Rule-Based

Trees follow rule-based paths.

2

Human-Readable

Easy to understand logic.

3

Mix-Friendly

Good with categorical and numerical mixes.

What the Model Learns from Variables

Binary Inputs

Parental help yes/no.

Ordinal Inputs

Health scores.

Continuous Inputs

Number of days.

Summary & Takeaways

Binary Classification:

•Tobacco use was best predicted by peer influence, parental support, and household income.
•The model accurately captured risk patterns in youth behavior using simple, interpretable splits.

Multi-Class Classification:

•Health status and peer context (such as friends' behavior) played a key role in predicting the frequency of substance use.
•The decision tree effectively differentiated between occasional and frequent users.

Regression:

•Though the model had lower predictive accuracy, it highlighted income level and social context as key factors influencing number of smoking days.
•Suggests room for improvement, possibly with more continuous variables or ensemble methods.

Overall Insight:

- Decision trees offered clarity: their rule-based nature helped us understand not just what the model predicted, but why.
- Ideal for public health applications where explainability is crucial for policymaking and intervention planning.

Thank You